

Equivariant and N_∞ -operads

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Main Take-Aways

Equivariantize All The Things!

Allow the group to act on any parameterizing diagrams.

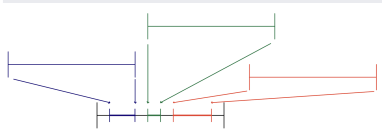
Internal & External Forms

Equivariant structures often have two equivalent forms:

{Mackey, Tambara, etc} versus G -commutative monoids.

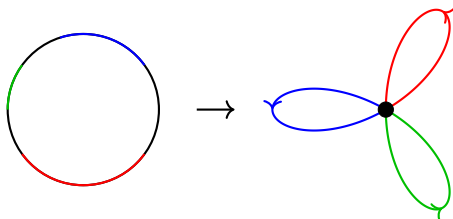
Little intervals

Embedding Intervals



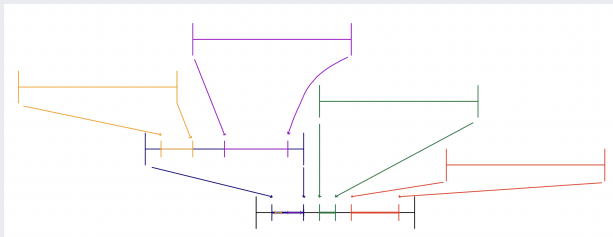
Look at the space

$$\mathcal{E}_1(n) = \text{Emb} \left(\coprod^n I, I \right).$$



Composition

Associativity



$$\mathcal{E}_1(3) \times (\mathcal{E}_1(2) \times \mathcal{E}_1(1) \times \mathcal{E}_1(1)) \rightarrow \mathcal{E}_1(4).$$

Permuting Coordinates

Have a Σ_n action on $\mathcal{E}_1(n)$

Operads & G-Operads

Definition (May)

An operad is

- 1 *a sequence of $G \times \Sigma_n$ -spaces $\mathcal{O}(n)$*
- 2 *an “identity” $i \in \mathcal{O}(1)$, and*
- 3 *composition maps*

$$\mathcal{O}(n) \times \mathcal{O}(k_1) \times \cdots \times \mathcal{O}(k_n) \rightarrow \mathcal{O}(k_1 + \cdots + k_n).$$

The identity is an identity for the composition, and the compositions associate and respect reordering the factors.

Operadic Algebras

Definition

An $\mathcal{O}$ -algebra is a space X with maps $\mathcal{O}(n) \times_{\Sigma_n} X^{\times n} \rightarrow X$ compatible with composition.

Example

The little disks in an orthogonal G -representation V :

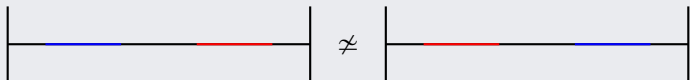
$$\mathcal{D}(V)_n = \text{Emb} \left(\coprod^n D(V), D(V) \right)$$

with the identity in $\mathcal{D}(V)_1$ as the identity, and with gluing in embeddings as composition.

Higher commutativity

Little intervals

The space $\mathcal{E}_1(2)$ is not path connected:



Little cubes

As k grows, the spaces

$$\mathcal{D}(\mathbb{R}^k)(n) = \text{Emb} \left(\coprod^n D^k, D^k \right)$$

become more and more connected.

Coherent Commutativity

 $\mathcal{E}_\infty$

An $\mathcal{E}_\infty$ -operad has $\mathcal{E}_\infty(n)$ a contractible, Σ_n -free space.

What does this give us?

Any choice of Σ_n -equivariant map $\Sigma_n \rightarrow \mathcal{E}_\infty(n)$ gives a multiplication

$$X^n \cong \Sigma_n \times_{\Sigma_n} X^n \rightarrow \mathcal{E}_\infty(n) \times_{\Sigma_n} X^n \rightarrow X.$$

Slogan

Ordinary $\mathcal{E}_\infty$ -operads structure “coherently commutative” ordinary multiplications, and there is only one, up to homotopy.

Non-trivial actions?

G -operads

What changes do non-trivial G -actions introduce?

Equivariant Little Disks

What happens for non-trivial representations?

Higher Commutativity

What plays the role of $\mathcal{E}_\infty$ here?

Elmendorf's Theorem Homotopy

Coefficient Systems of Spaces

Given a G -space X , we have a functor $\underline{X}: \mathcal{O}rb^{G,op} \rightarrow \mathcal{T}op$:

$$T \mapsto \text{Map}^G(T, X).$$

Theorem (Elmendorf)

The assignment $X \mapsto \underline{X}$ gives an equivalence of infinity categories

$$\mathcal{T}op^G \simeq \text{Fun}(\mathcal{O}rb^{G,op}, \mathcal{T}op).$$

Rewriting $\mathcal{D}(V)$

Equivariant Homotopy Type

Evaluation at the origins gives an equivalence

$$\mathcal{D}(V)(n) \xrightarrow{\cong} \text{Emb}(\{1, \dots, n\}, D(V)) \cong \text{Conf}_n(V).$$

Action

An element $(g, \varsigma) \in G \times \Sigma_n$ acts on an embedding $f: \{1, \dots, n\} \rightarrow V$ as

$$(g, \varsigma) \cdot f(j) = g\left(f(\varsigma^{-1}(j))\right).$$

Fixed points of $\mathcal{D}(V)$

Proposition

Σ_n acts freely on $\mathcal{D}(V)(n)$.

Proposition

If $\Gamma \subset G \times \Sigma_n$ has $\Gamma \cap \Sigma_n = \{e\}$, then

$$\Gamma = \left\{ (h, f(h)) \mid h \in H \subseteq G, f: H \rightarrow \Sigma_n \right\}.$$

Graph Subgroups

Graph Subgroups of $G \times \Sigma_n$ are the same data as a subgroup H and an H -set structure on $\{1, \dots, n\}$.

Fixed points of $\mathcal{D}(V)$

Theorem

Let Γ be a graph subgroup corresponding to a finite H -set T of cardinality n . Then

$$\mathrm{Conf}_n(V)^\Gamma \cong \mathrm{Emb}^H(T, V).$$

Corollary

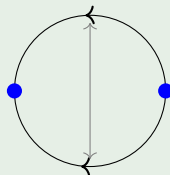
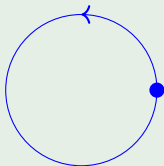
Let X be a $\mathcal{D}(V)$ -algebra. Then we have $\mathrm{Emb}^H(T, V)$ -maps

$$\mathrm{Map}(T, i_H^* X) \rightarrow i_H^* X.$$

C_2 -equivariant 1-spheres

Example

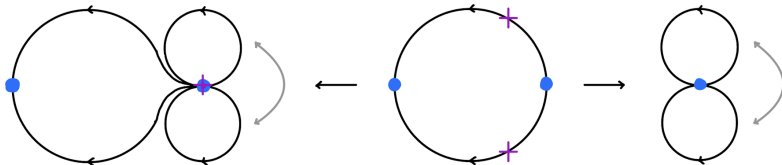
If $G = C_2$, then we have S^1 and S^σ , the unit complex numbers with complex conjugation.



Two kinds of loop spaces

Have $\Omega X = \text{Map}_*(S^1, X)$ and $\Omega^\sigma X = \text{Map}_*(S^\sigma, X)$.

Signed loop spaces



“Multiplications”

$$\Omega^\sigma X \times (\underbrace{\Omega X \times \Omega X}) \rightarrow \Omega^\sigma X \leftarrow (\underbrace{\Omega X \times \Omega X}).$$

No ordinary product

$\Omega^\sigma X$ does not have a multiplication! $\Omega^\sigma X \times \Omega^\sigma X \nrightarrow \Omega^\sigma X$

Structure on signed loops

Underlying

- Non-equivariantly ΩX is an ordinary loop-space
- C_2 acts as an anti-automorphism: the inverse

Fixed Points

- Have an action of ΩX on $(\Omega^\sigma X)^{C_2}$, and
- Have a “unit” map $\Omega X \rightarrow (\Omega^\sigma X)^{C_2}$.

Mackey Functors

The signed-loop action gives us a transfer map!

N_∞ Operads

Definition (Blumberg–H.)

An N_∞ operad for G is a G -operad $\mathcal{O}(n)$ such that

- ① Σ_n -acts freely,
- ② if $\Gamma \subseteq G \times \Sigma_n$, the fixed points $\mathcal{O}(n)^\Gamma$ are either empty or contractible, and
- ③ the G -fixed points $\mathcal{O}(n)^G$ are always contractible.

Example

If U is a universe, then

- ① $\mathcal{D}(U)$ is N_∞
- ② the linear isometries operad $\mathcal{L}(U)$ is N_∞

Rewriting: Admissible Sets

Definition

A finite H -set T is admissible for $\mathcal{O}$ if $\mathcal{O}(|T|)^{\Gamma_T} \neq \emptyset$ for Γ_T any graph subgroup classifying the H -set T .

Example

The finite G -set $* \amalg \cdots \amalg *$ is admissible for any $\mathcal{O}$.

Example

T is admissible for $\mathcal{D}(U)$ iff $T \hookrightarrow U$.

Example

T is admissible for $\mathcal{L}(U)$ iff $\mathbb{R} \cdot T \otimes U \hookrightarrow U$.

N_∞ operads do the right thing

Theorem (Blumberg–H.)

If T is an admissible H -set for $\mathcal{O}$, then $\mathcal{O}$ structures an essentially unique H -equivariant twisted multiplication

$$\mathrm{Map}(T, X) \rightarrow X.$$

Example

Little disks for $\infty\rho$. We have all transfers

$$\mathrm{Map}(T, X) \rightarrow X.$$

Combinatorial implications

Interconnected Transfers

T being admissible for H implies the existence of lots of other admissible sets

Example

The transfer

$$\mathrm{Map}(C_2, \Omega X) \rightarrow \Omega^\sigma X$$

gave the ordinary loop multiplication.

Theorem

The admissible sets are a complete invariant of N_∞ operads.

Indexing Systems

Definition (Blumberg–H.)

An indexing system is a collection of subcategories $\underline{\mathcal{Q}}(H) \subseteq \mathcal{F}in^H$, one for each subgroup $H \subseteq G$ such that

- ① *$\underline{\mathcal{Q}}(H)$ is closed under disjoint unions and summands,*
- ② *if $K \subseteq H$, then $T \in \underline{\mathcal{Q}}(H)$ implies $i_K^* T \in \underline{\mathcal{Q}}(K)$,*
- ③ *if $g \in G$, then $T \in \underline{\mathcal{Q}}(H)$ implies $c_g^* T \in \underline{\mathcal{Q}}(gHg^{-1})$,*
- ④ *if $H/K \in \underline{\mathcal{Q}}(H)$, then $T \in \underline{\mathcal{Q}}(K)$ implies $H \times_K (T) \in \underline{\mathcal{Q}}(H)$.*

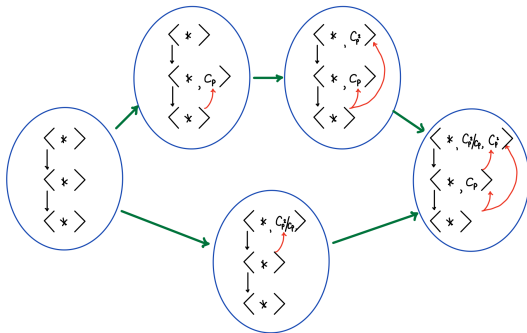
Poset

The admissible H -sets as H varies forms an indexing system.

Poset of Indexing Systems

The Poset $\mathcal{I}(G)$

$\underline{\mathcal{O}} \leq \underline{\mathcal{O}}'$ if for all H , $\underline{\mathcal{O}}(H) \subseteq \underline{\mathcal{O}}'(H)$.



Transfer Systems

Definition (Rubin, Balchin–Barnes–Roitzheim)

A transfer system on G is a poset structure $\rightarrow$ on $\text{Sub}(G)$ satisfying

- 1 If $K \rightarrow H$, then $K \subset H$,
- 2 if $K \rightarrow H$ and $g \in G$, then $gKg^{-1} \rightarrow gHg^{-1}$, and
- 3 if $K \rightarrow H$ and $J \subset H$, then $K \cap J \rightarrow J$.

Theorem (Rubin, Balchin–Barnes–Roitzheim)

$$\text{Tran}(G) \cong \mathcal{I}(G).$$

Classification Theorem

Theorem (Blumberg–H.)

$\mathcal{O} \mapsto \underline{\mathcal{O}}$ gives a fully faithful embedding of ∞ -categories

$$\mathcal{N}_\infty^G \hookrightarrow \mathcal{I}(G).$$

Theorem (Bonventre–Pereira, Gutiérrez–White, Rubin)

Every indexing system is the indexing system associated to an N_∞ -operad.